OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

6.5 Overview of elements

Table 12. Overview of ASAM OpenDRIVE elements

Element	Section
`<OpenDRIVE>`	
| — `<header>`	Section 6.4.1, " `<header>` element"
| | — `<geoReference>`	Section 8.5, "Georeferencing"
| | — `<offset>`	Section 8.5, "Georeferencing"
| | — `<license>`	Section 6.4.2, " `<license>` element"
| | — `<defaultRegulations>`	Section 6.4.3, " `<defaultRegulations>` element"
| | | — `<roadRegulations>`	Section 6.4.3, " `<defaultRegulations>` element"
| | | — `<signalRegulations>`	Section 6.4.3, " `<defaultRegulations>` element"
| — `<road>`	Section 10.1, "Introduction to roads"
| | — `<link>`	Section 10.3, "Road linkage"
| | | — `<predecessor>`	Section 10.3, "Road linkage"
| | | — `<successor>`	Section 10.3, "Road linkage"
| | — `<type>`	Section 10.4, "Road type"
| | | — `<speed>`	Section 10.4.1, "Speed limits for road types"

Element	Section
| | — <planView>	Section 12.2, "only_one_line_element"
| | | — <geometry>	Section 12.9, "only_one_line_element"
| | | | — <line>	Section 9.3, "Straight line"
| | | | — <spiral>	Section 9.4, "Spiral"
| | | | — <arc>	Section 9.5, "Arc"
| | | | — <poly3>	Section 9.7, "Cubic polynom (deprecated)"
| | | | — <paramPoly3>	Section 9.6, "Parametric cubic curve"
| | — <elevationProfile>	Section 10.5.1, "Road elevation"
| | | — <elevation>	Section 10.5.1, "Road elevation"
| | — <lateralProfile>	Section 10.5.1, "Superelevation"
| | | — <superelevation>	Section 10.5.1, "Superelevation"
| | | — <shape>	Section 10.5.1, "Shape definition"
| | | — <crossSectionSurface>	Section 10.5.1, "Cross section surface"
| | | | — <tOffset>	Section 10.5.1, "Cross section surface"
| | | | | — <coefficients>	Section 10.5.1, "Cross section surface"
| | | | — <surfaceStrips>	Section 10.5.1, "Cross section surface"
| | | | | — <strip>	Section 10.5.1, "Cross section surface"
| | | | | | — <width>	Section 10.5.1, "Cross section surface"

Element	Section
| | | | | | | — <coefficients>	Section 10.5.1, "Cross section surface"
| | | | | | — <constant>	Section 10.5.1, "Cross section surface"
| | | | | | | — <coefficients>	Section 10.5.1, "Cross section surface"
| | | | | | — <linear>	Section 10.5.1, "Cross section surface"
| | | | | | — <quadratic>	Section 10.5.1, "Cross section surface"
| | | | | | | — <coefficients>	Section 10.5.1, "Cross section surface"
| | | | | | — <cubic>	Section 10.5.1, "Cross section surface"
| | | | | | | — <coefficients>	Section 10.5.1, "Cross section surface"
| | — <lanes>	Section 11.1, "Introduction to lanes"
| | | — <laneOffset>	Section 11.4, "Lane offset"
| | | — <laneSection>	Section 11.3, "Lane sections"
| | | | — <left>	Section 11.2, "Lane groups"
| | | | | — <lane>	Section 11.2, "Lane groups"
| | | | | | — <width>	Section 11.6.1, "Lane width"
| | | | | | — <border>	Section 11.6.1, "Lane borders"
| | | | | | — <link>	Section 11.5, "Lane linkage"
| | | | | | | — <predecessor>	Section 11.5, "Lane linkage"
| | | | | | | — <successor>	Section 11.5, "Lane linkage"

Element	Section
| | | | | | — <roadMark>	Section 11.8, "Road markings"
| | | | | | | — <sway>	Section 11.8.3, "Offset in road markings"
| | | | | | | — <type>	Section 11.8.1, "Road marking types and lines"
| | | | | | | | — <line>	Section 11.8.1, "Road marking types and lines"
| | | | | | | — <explicit>	Section 11.8.2, "Explicit road marking types and lines"
| | | | | | | | — <line>	Section 11.8.2, "Explicit road marking types and lines"
| | | | | | — <material>	Section 11.7.2, "Lane material"
| | | | | | — <speed>	Section 11.7.1, "Lane speed limit"
| | | | | | — <access>	Section 11.7.1, "Lane access"
| | | | | | | — <restriction>	Section 11.7.1, "Lane access"
| | | | | | — <height>	Section 11.6.1, "Lane height"
| | | | | | — <rule>	Section 11.9, "Specific lane rules"
| | | | —	Section 11.2, "Lane groups"
| | | | | — <lane>	Section 11.2, "Lane groups"
| | | | | | — <link>	Section 11.5, "Lane linkage"
| | | | | | | — <predecessor>	Section 11.5, "Lane linkage"
| | | | | | | — <successor>	Section 11.5, "Lane linkage"
| | | | | | — <roadMark>	Section 11.8, "Road markings"

Element	Section
| | | | | | | — <sway>	Section 11.8.3, "Offset in road markings"
| | | | | | | — <type>	Section 11.8.1, "Road marking types and lines"
| | | | | | | | — <line>	Section 11.8.1, "Road marking types and lines"
| | | | | | | — <explicit>	Section 11.8.2, "Explicit road marking types and lines"
| | | | | | | | — <line>	Section 11.8.2, "Explicit road marking types and lines"
| | | | — <right>	Section 11.2, "Lane groups"
| | | | | — <lane>	Section 11.2, "Lane groups"
| | | | | | — <width>	Section 11.6.1, "Lane width"
| | | | | | — <border>	Section 11.6.1, "Lane borders"
| | | | | | — <link>	Section 11.5, "Lane linkage"
| | | | | | | — <predecessor>	Section 11.5, "Lane linkage"
| | | | | | | — <successor>	Section 11.5, "Lane linkage"
| | | | | | — <roadMark>	Section 11.8, "Road markings"
| | | | | | | — <sway>	Section 11.8.3, "Offset in road markings"
| | | | | | | — <type>	Section 11.8.1, "Road marking types and lines"
| | | | | | | | — <line>	Section 11.8.1, "Road marking types and lines"
| | | | | | | — <explicit>	Section 11.8.2, "Explicit road marking types and lines"
| | | | | | | | — <line>	Section 11.8.2, "Explicit road marking types and lines"

Element	Section
| | | | | | — <material>	Section 11.7.2, "Lane material"
| | | | | | — <speed>	Section 11.7.1, "Lane speed limit"
| | | | | | — <access>	Section 11.7.1, "Lane access"
| | | | | | | — <restriction>	Section 11.7.1, "Lane access"
| | | | | | — <height>	Section 11.6.1, "Lane height"
| | | | | | — <rule>	Section 11.9, "Specific lane rules"
| | — <objects>	Section 13.1, "Introduction to objects"
| | | — <object>	Section 13.1, "Introduction to objects"
| | | | — <repeat>	Section 13.2, "Repeating objects"
| | | | — <outline>	Section 13.3, "Object outline"
| | | | | — <cornerRoad>	Section 13.3.1, "<cornerRoad> element"
| | | | | — <cornerLocal>	Section 13.3.1, "<cornerLocal> element"
| | | | — <outlines>	Section 13.3, "Object outline"
| | | | | — <outline>	Section 13.3, "Object outline"
| | | | | | — <cornerRoad>	Section 13.3.1, "<cornerRoad> element"
| | | | | | — <cornerLocal>	Section 13.3.1, "<cornerLocal> element"
| | | | — <material>	Section 13.5, "Object material"
| | | | — <validity>	Section 14.2, "Lane validity for signals"

Element	Section
| | | | — <parkingSpace>	Section 13.7, "Access rules to parking spaces"
| | | | — <markings>	Section 13.8, "Object marking"
| | | | | — <marking>	Section 13.8, "Object marking"
| | | | | | — <cornerReference>	Section 13.8, "Object marking"
| | | | — <borders>	Section 13.9, "Object borders"
| | | | | — <border>	Section 13.9, "Object borders"
| | | | | | — <cornerReference>	Section 13.8, "Object marking"
| | | | — <surface>	Section 13.13, "Object CRG surface"
| | | | — <skeleton>	Section 13.4, "Object skeleton"
| | | | | — <polyline>	Section 13.4, "Object skeleton"
| | | | | | — <vertexRoad>	Section 13.4.1, " <vertexRoad> element"
| | | | | | — <vertexLocal>	Section 13.4.1, " <vertexLocal> element"
| | | — <objectReference>	Section 13.10, "Object reference"
| | | | — <validity>	Section 14.2, "Lane validity for signals"
| | | — <tunnel>	Section 13.11, "Tunnels"
| | | | — <validity>	Section 14.2, "Lane validity for signals"
| | | — <bridge>	Section 13.12, "Bridges"
| | | | — <validity>	Section 14.2, "Lane validity for signals"

Element	Section
| | — <signals>	Section 14.1, "Introduction to signals"
| | | — <signal>	Section 14.1, "Introduction to signals"
| | | | — <positionInertial>	Section 14.9, "Signal positioning (deprecated)"
| | | | — <positionRoad>	Section 14.9, "Signal positioning (deprecated)"
| | | | — <validity>	Section 14.2, "Lane validity for signals"
| | | | — <dependency>	Section 14.3, "Signal dependency"
| | | | — <reference>	Section 14.4, "Signal reference"
| | | — <signalReference>	Section 14.5, "Signals that apply to multiple roads"
| | | | — <validity>	Section 14.2, "Lane validity for signals"
| | — <surface>	Section 12.12, "Junction CRG surface"
| | | — <CRG>	Section 12.12, "Junction CRG surface"
| | — <railroad>	Section 15.1, "Introduction to railroads"
| | | — <switch>	Section 15.3, "Switches"
| | | | — <mainTrack>	Section 15.3.1, "Main track"
| | | | — <sideTrack>	Section 15.3.1, "Side track"
| | | | — <partner>	Section 15.3.2, "Partner switches"
| — <controller>	Section 14.6, "Signal Controllers"
| | — <control>	Section 14.6, "Signal Controllers"

Element	Section
| — <junction type="crossing">	Section 12.8, "Crossings"
| | — <roadSection>	Section 12.8, "Crossings"
| | — <priority>	Section 12.4.1, "Priorities of connecting roads within a junction"
| | — <controller>	Section 12.14, "Signal synchronization groups in junctions"
| | — <surface>	Section 12.12, "Junction CRG surface"
| | | — <CRG>	Section 12.12, "Junction CRG surface"
| | — <planView>	Section 12.2, "only_one_line_element"
| | | — <geometry>	Section 12.9, "only_one_line_element"
| | | | — <line>	Section 9.3, "Straight line"
| | | | — <spiral>	Section 9.4, "Spiral"
| | | | — <arc>	Section 9.5, "Arc"
| | | | — <poly3>	Section 9.7, "Cubic polynom (deprecated)"
| | | | — <paramPoly3>	Section 9.6, "Parametric cubic curve"
| — <junction type="default">	Section 12.2, "when_to_use"
| | — <connection>	Section 12.4, "Connecting roads"
| | | — <laneLink>	Section 12.6, "Direct junctions"
| | — <crossPath>	Section 12.5, "Cross paths"
| | | — <startLaneLink>	Section 12.5, "Cross paths"

Element	Section
| | | — <endLaneLink>	Section 12.5, "Cross paths"
| | — <priority>	Section 12.4.1, "Priorities of connecting roads within a junction"
| | — <controller>	Section 12.14, "Signal synchronization groups in junctions"
| | — <surface>	Section 12.12, "Junction CRG surface"
| | | — <CRG>	Section 12.12, "Junction CRG surface"
| | — <planView>	Section 12.2, "only_one_line_element"
| | | — <geometry>	Section 12.9, "only_one_line_element"
| | | | — <line>	Section 9.3, "Straight line"
| | | | — <spiral>	Section 9.4, "Spiral"
| | | | — <arc>	Section 9.5, "Arc"
| | | | — <poly3>	Section 9.7, "Cubic polynom (deprecated)"
| | | | — <paramPoly3>	Section 9.6, "Parametric cubic curve"
| | — <boundary>	Section 12.10, "Junction boundary"
| | | —	Section 12.10, "Junction boundary"
| | | —	Section 12.10, "Junction boundary"
| | — <elevationGrid>	Section 12.11, "Junction elevation grid"
| | | — <elevation>	Section 12.11, "Junction elevation grid"
| — <junction type="direct">	Section 12.6, "Direct junctions"

Element	Section
| | — <connection>	Section 12.6, "Direct junctions"
| | | — <laneLink>	Section 12.6, "Direct junctions"
| | — <priority>	Section 12.4.1, "Priorities of connecting roads within a junction"
| | — <controller>	Section 12.14, "Signal synchronization groups in junctions"
| | — <surface>	Section 12.12, "Junction CRG surface"
| | | — <CRG>	Section 12.12, "Junction CRG surface"
| | — <planView>	Section 12.2, "only_one_line_element"
| | | — <geometry>	Section 12.9, "only_one_line_element"
| | | | — <line>	Section 9.3, "Straight line"
| | | | — <spiral>	Section 9.4, "Spiral"
| | | | — <arc>	Section 9.5, "Arc"
| | | | — <poly3>	Section 9.7, "Cubic polynom (deprecated)"
| | | | — <paramPoly3>	Section 9.6, "Parametric cubic curve"
| — <junction type="virtual">	Section 12.7, "Virtual junctions"
| | — <connection type="default">	Section 12.7.1, "Cross paths with virtual junctions"
| | | — <laneLink>	Section 12.6, "Direct junctions"
| | — <connection type="virtual">	Section 12.7.1, "Virtual connections (deprecated)"
| | | — <predecessor>	Section 12.7.1, "Virtual connections (deprecated)"

Element	Section
| | | — <successor>	Section 12.7.1, "Virtual connections (deprecated)"
| | | — <laneLink>	Section 12.6, "Direct junctions"
| | — <crossPath>	Section 12.5, "Cross paths"
| | | — <startLaneLink>	Section 12.5, "Cross paths"
| | | — <endLaneLink>	Section 12.5, "Cross paths"
| | — <priority>	Section 12.4.1, "Priorities of connecting roads within a junction"
| | — <controller>	Section 12.14, "Signal synchronization groups in junctions"
| | — <surface>	Section 12.12, "Junction CRG surface"
| | | — <CRG>	Section 12.12, "Junction CRG surface"
| | — <planView>	Section 12.2, "only_one_line_element"
| | | — <geometry>	Section 12.9, "only_one_line_element"
| | | | — <line>	Section 9.3, "Straight line"
| | | | — <spiral>	Section 9.4, "Spiral"
| | | | — <arc>	Section 9.5, "Arc"
| | | | — <poly3>	Section 9.7, "Cubic polynom (deprecated)"
| | | | — <paramPoly3>	Section 9.6, "Parametric cubic curve"
| — <junctionGroup>	Section 12.13, "Junction groups"
| | — <junctionReference>	Section 12.13, "Junction groups"

Element	Section
| — `<station>`	Section 15.4, "Stations"
| | — `<platform>`	Section 15.4.1, "Platforms"
| | | — `<segment>`	Section 15.4.1, "Segments"